Topic Test 1 · Paper 1

Question 1

The expansion of $(a - bx)^c$ is

$$4 - px + 108x^2 - qx^3 + rx^4$$

where a, b, c, p, q, r are positive real constants.

Find the value of $p + q + r$.

A $81 - 84\sqrt{2}$

B $81 + 132\sqrt{2}$

C $132\sqrt{2} - 81$

D $81 + 84\sqrt{2}$

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Question 2

The coefficient of x^{11} in $(2+x^2+x^3)^8$ is equal to 28 times the coefficient of x^2 in $(2+ax)^6$.

Find all possible values of a.

A $\pm 2\sqrt{7}$

B $\pm \frac{1}{2}$

C $\pm \frac{1}{4}$

D ± 1

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Question 3

The coefficient of x^2 in $(2+bx)^4$ is 2 times the coefficient of x^3 in $(1+bx)^6$.

Given $b \neq 0$, find b.

A $\frac{8}{15}$

B $\frac{3}{5}$

C $\frac{3}{20}$

D $\frac{6}{5}$

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Question 4

Find the coefficient of x in

$$(1-x)^0-(1-x)^1+(1-x)^2-(1-x)^3+\cdots+(1-x)^{50}.$$

A 51

B -51

C -25

D 25

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Question 5

Find the value of

$$\sqrt{16+8\sqrt{3}+3}-\sqrt{12-4\sqrt{3}+1}.$$

A $\sqrt{6+12\sqrt{3}}$

B $\sqrt{3}-5$

C 3

D $5-\sqrt{3}$

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Question 6

A movable circle of radius 3 touches the x-axis from above. A fixed circle has centre at the origin and radius 8. When the two circles touch, the movable centre has coordinates $(t, 3)$. What is the sum of t^2 over all distinct possible values of t?

A 128

B 160

C 224

D 256

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Question 7

Find the value(s) of a such that the turning point of

$$y = x^2 + 2ax + 1$$

is closest to the origin.

A $a = \sqrt{2}$

B $a = \pm\sqrt{2}$

C $a = \frac{\sqrt{2}}{2}$

D $a = \pm\frac{1}{\sqrt{2}}$

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Question 8

The line $y = 2x + c$ intersects the circle $x^2 + y^2 = 9$ at two points A and B. M is the midpoint of chord AB.

The equation of the locus of M, as c varies between $-3\sqrt{5}$ and $3\sqrt{5}$, is:

A $x^2 + y^2 = 2$

B $y = -\frac{1}{2}x$

C $y = \frac{1}{2}x$

D $y = x^2$

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Question 9

The tangent to

$$x^2 + y^2 + 6x + 2y + 2 = 0$$

at $(-5, 1)$ passes through $(-3, 3)$. The other tangent from $(-3, 3)$ touches the circle at:

A $(-3, 2\sqrt{2})$

B $(-3, 1)$

C $(-1, 2\sqrt{2})$

D $(-1, 1)$

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Question 10

The circle

$$x^2 + y^2 + 2ax + 2by = c$$

encloses $(1, 1)$ precisely when:

A $c > 2(1 - a - b)$

B $a^2 + b^2 < 1$

C $c > 2(a + b + 1)$

D $a^2 + 2ab + b^2 < c$

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Question 11

Which of the following is a tangent to the circle

$$(x-2)^2+(y-3)^2=4?$$

A $x=3+\sqrt{2}$

B $y=5+\frac{\sqrt{2}}{2}-x$

C $y=2(2-\sqrt{2})$

D $y=x+1-2\sqrt{2}$

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Question 12

The reflection of (a, a) in the line

$$y = \frac{1}{a}x, \qquad a \neq 0$$

is:

A $\left(\frac{a(a-1)^2}{a^2+1}, \frac{a^2(a^2+3)}{a^2+1}\right)$

B $\left(\frac{a(a^2+2a-1)}{a^2+1}, \frac{-a(a^2-2a-1)}{a^2+1}\right)$

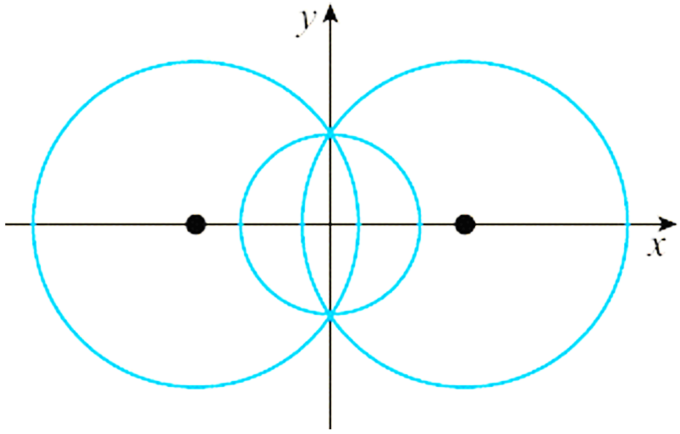
C $\left(\frac{a(a-1)^2}{a^2+1}, \frac{-a(a-1)^2}{a^2+1}\right)$

D $\left(\frac{a(a+1)^2}{a^2+1}, \frac{a^2(a+1)^2}{a^2+1}\right)$

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Question 13

The circle C has equation $x^2 + y^2 = 1$. Two circles of equal radius r, with centres $(a, 0)$ and $(-a, 0)$, where $a > 0$, both pass through $(0, 1)$ and $(0, -1)$. This is shown in the diagram.



The two arcs divide C into three distinct regions. If the three regions have equal area, which equation is satisfied by a?

A $3(a^2 + 1)\arctan\left(\frac{1}{a}\right) - 3a = \pi$

B $6(a^2 + 1)\arctan\left(\frac{1}{a}\right) - 6a = \pi$

C $6(a^2 + 1)\arctan\left(\frac{1}{a}\right) - 3a = \pi$

D $12(a^2 + 1)\arctan\left(\frac{1}{a}\right) - 6a = \pi$

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Question 14

A sequence starts at $u_1 = 1$. To obtain u_{n+1}, add 3 to u_n if n is odd, and halve u_n if n is even. What is the least odd index n for which $u_n > 23/8$?

A 7

B 8

C 9

D 11

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Question 15

A polynomial P leaves remainder 5 on division by $x+1$ and remainder -1 on division by $x-2$. What is the remainder when $P(x)+P(1-x)$ is divided by $(x+1)(x-2)$?

A $2x+1$

B 6

C 4

D $4x$

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Question 16

For a positive integer n, the equation $3^{2x} - (n+2)3^x + 2n = 0$ has two real roots that differ by exactly 2. Find n.

A 18

B 6

C 9

D 12

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Question 17

Let n be a positive integer. $x^2 + 2$ is a factor of

$$(x^4 - 5)^n - (x^2 + 1)^{n+1} + (x^2 + 4)^n (x^2 + 3)^n$$

for:

A odd n

B even n

C no n

D all n

E $n = 1$

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Question 18

A monic cubic P satisfies $P(0) = 0$ and $P(x+1) - P(x) = 3x^2 - 11x + 4$ for every real x. What remainder does $P(x)$ leave on division by $x + 1$?

A -24

B -20

C -18

D -16

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Question 19

The polynomial p_n is defined by

$$p_n(x) = (x - 2n + 1) + (2x - 2n + 3) + (3x - 2n + 5) + \cdots + (nx - 1).$$

Given $n \geq 2$, what is the remainder when $p_n(x)$ is divided by $p_{n-1}(x)$?

A $\frac{n^2-1}{2}$

B n

C -1

D 1

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Question 20

Given that x, y, z are positive real numbers,

$$4\log_z x = y, \qquad \log_x z = y, \qquad x + \log_y z = 0.$$

Which of the following is true?

A Unique solution for x, but not for y and z.

B Unique solutions for x and y, but infinitely many solutions for z.

C Unique solutions for x and z, but infinitely many solutions for y.

D Unique solution for x, y and z.